



Resources allowed: A handwritten A4 notes page double sided
 Calculator

Name: Answer Key Class: _____ Total Time: 55 minutes Mark /54

Question 1**2 marks**

A taxi departs at 18:32 from Baldivis and arrives at 20:13 at Armadale. How long did the taxi take for the trip?

1hr 41min
 ✓ ✓

Question 2**3 marks**

Convert the following

a) 180 sec = 3 min ✓

b) 3 days = 4320 min ✓

c) 4 weeks = 672 hours ✓

Question 3**3 marks**

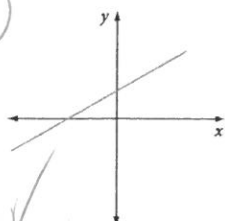
Jarrod went to the movies with some friends. If the movie started at 6:45pm and the movie went for 1 hour and 30 mins, at time did the movie finish?

a) In 12 hour time 8:15pm b) In 24 hour time 2015
 ✓ ✓

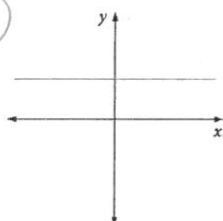
Question 4**4 marks**

Which of these graphs show a linear relationship (Circle)?

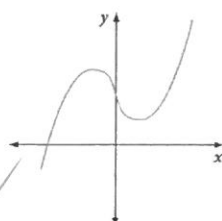
a



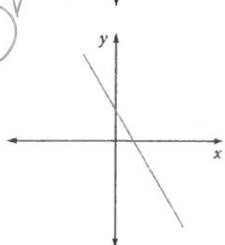
b



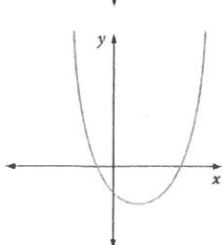
c



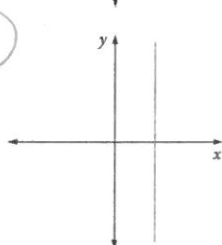
d



e

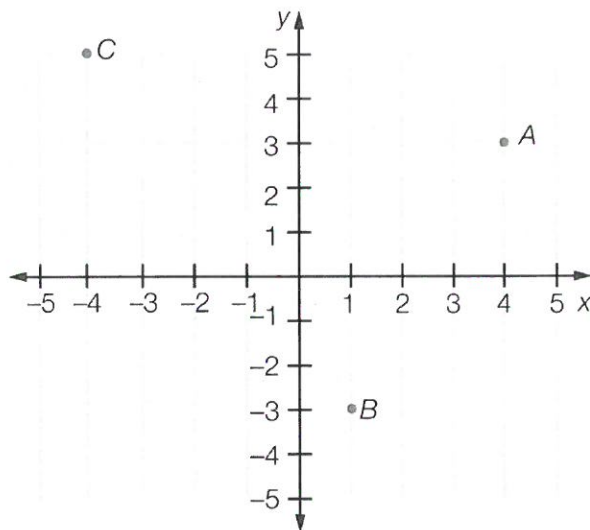


f



Question 5
3 marks

Look at the Cartesian plane below and write down the coordinates of the points A, B and C.

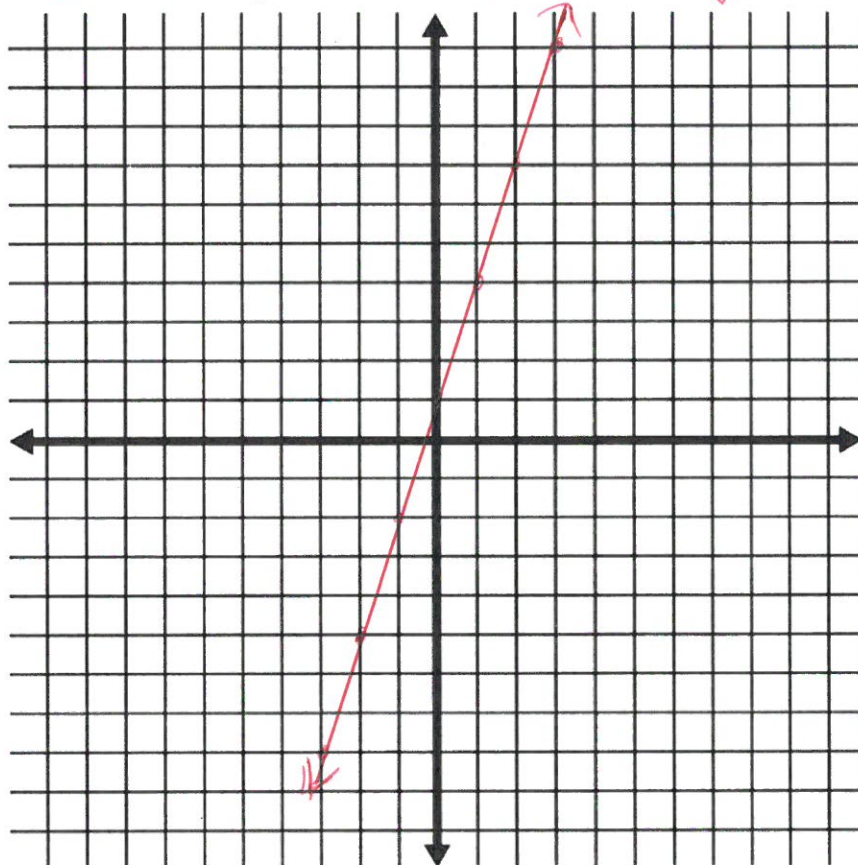


A (4, 3) ✓
 B (1, -3) ✓
 C (-4, 5) ✓

Question 6
10 marks

Use the rule $y = 3x + 1$ to fill in the table below then plot as a linear graph

x	-3	-2	-1	0	1	2	3
y	7	5	2	1	4	7	10



4 marks for
 correct 6 coords.
 1 mark for line
 (straight)
 1 mark for
 arrows on line

Question 7**2 marks**

The x-coordinate of a point on the line $y = 10 - 3x$ is 3. Write down the y-coordinate.

$$\begin{aligned}y &= 10 - 3(3) \\y &= 10 - 9 \checkmark \\y &= 1 \checkmark\end{aligned}$$

Question 8**4 marks**

Find the rule for these tables of values.

a

x	-2	-1	0	1	2
y	8	5	2	-1	-4

$$y = -3x + 2 \checkmark \checkmark$$

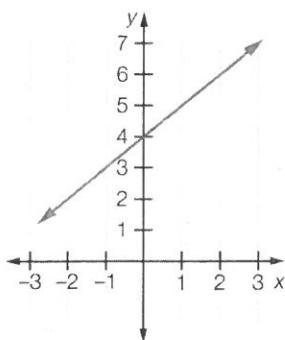
b

x	2	3	4	5	6
y	-1	1	3	5	7

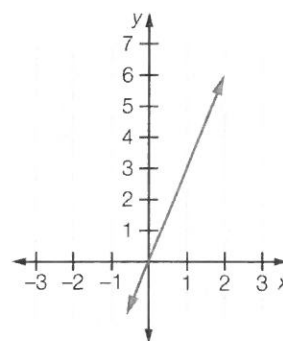
$$y = 2x - 5 \checkmark \checkmark$$

Question 9**4 marks**

Write down the equation of each of the following lines.

a)

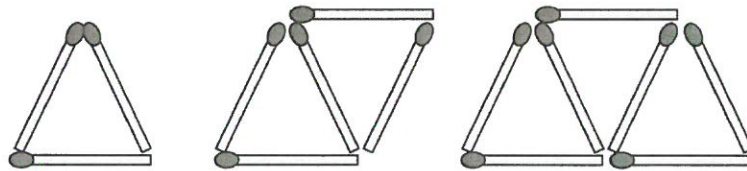
$$y = x + 4 \checkmark \checkmark$$

b)

$$y = 3x \checkmark \checkmark$$

Question 10
6 marks

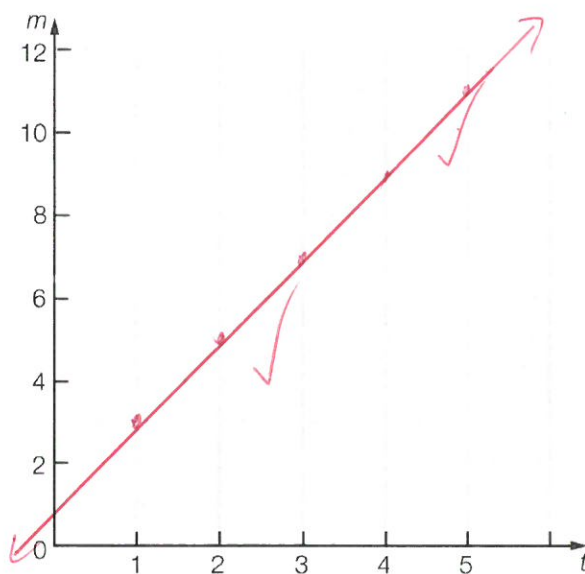
The following shapes have been made with matchsticks



a) Complete the table of values.

Number of triangles (t)	Number of matches (m)
1	3
2	5
3	7
4	9
5	11

b) Graph each set of values and find the number of matches needed to build 20 triangles.



c) Find the pattern, or rule, in terms of m and t .

$$m = 2t + 1$$

$$m = 2(20) + 1$$

$$m = 41$$

Question 11
3 marks

$\frac{1}{2}$ each. D Probability Experiment
A Outcome
B Sample Space
E Event
F Complement
C Trial

A: A possible result.
 B: All possible results.
 C: An activity that involves chance.
 D: Doing the activity.
 E: A particular result.
 F: Not a particular result.

Question 12
10 marks

The two-way table below shows the results of a poll conducted of a group of students who own mobile phones to see who pays their own bills.

	Boys	Girls	Total
Pay own bill	7	16 ✓	23
Do not pay own bill	5 ✓	4	9 ✓
Total	12	20	32 ✓

- a) Complete the above two-way table
- b) How many people participated in this poll?
- 32 ✓
- c) How many of the students surveyed did not pay their own bill?
- 9 ✓

- d) Find the **probability** that a randomly selected student:

i. Is a girl who pays her own bill

$$\frac{16}{20} = \frac{4}{5} \checkmark$$

ii. Does not pay their own bill

$$\frac{9}{32} \checkmark \checkmark$$

END OF TEST